

TOPICAL REVIEW

Utility-Weighted Analysis of the Modified Rankin Scale: An Updated Review of Methods, Challenges, and Future Directions

Paul M. Wechsler¹ MD; Joseph P. Broderick² MD; Jeffrey L. Saver³ MD; Eva A. Mistry⁴ MBBS, MSCI

ABSTRACT: The utility-weighted approach to analyzing the modified Rankin Scale (mRS) is increasingly being utilized in acute stroke clinical trials. This analytic approach assigns a patient-centered utility weight—the desirability or value of a health state to patients—to each level of the mRS. The utility-weighted-mRS analysis converts the mRS from a scale reflecting only the rank order of functional outcome after stroke into one that quantifies the value of each poststroke mRS health state from patients' perspectives. The ability of the utility-weighted-mRS analysis to capture the unequal difference in patient health-related quality of life between each level of the mRS makes it an appealing patient-centered end point for acute stroke clinical trials. However, several concerns have historically raised skepticism regarding its value as a trial end point. Unfamiliarity with how utility weights are derived and used poses challenges to its use for shared decision-making between patients and clinicians. There are valid concerns that social, geographic, and demographic factors that differ between countries influence the mRS utility weights, so that use of uniform weights across all settings mildly reduces the precision of utility quantification, particularly in multinational stroke clinical trials. This narrative review aims to provide the clinical and research stroke community with a practical overview of key methodology in the utility-weighted-mRS analysis to aid in its interpretation, considers the advantages and challenges of using the utility-weighted-mRS analysis, and suggests future areas of study.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: clinical trial ■ demography ■ economics ■ policy ■ population health

The modified Rankin Scale (mRS) is the most common end point used in acute stroke clinical trials. It is a scale that measures a patient's global functional status, ranging from 0 (no symptoms) to 6 (death), with scores 1 to 5 describing increasing levels of global functional disability. Acute stroke clinical trials employ various approaches to analyzing the mRS. The dichotomized analytic approach examines whether an intervention improves outcomes across only one particular health state transition of the mRS scale, such as living independently or better (mRS score, 0–2) versus being dependent on others or dead (mRS score, 3–6).¹ The ordinal analytic approach examines how an intervention changes the distribution of patients' functional statuses after stroke across all levels of the mRS (mRS

score, 0–6, with mRS score of 5 and 6 sometimes combined into 1 level). The dichotomized and ordinal analyses of the mRS are the most used approaches in acute stroke clinical trials. However, the utility-weighted-mRS (UW-mRS) analysis is increasingly being used as the primary or important secondary analytic method for assessing the mRS in acute stroke clinical trials.^{2,3} In this approach, a patient-centered utility weight—the desirability or value of a health state to patients—is assigned to each level of the mRS. By capturing the value to patients of each mRS functional level, the UW-mRS analysis converts the mRS from a scale reflecting only the rank order of functional outcomes into a scale that quantifies the valuation of each mRS functional status from patients' perspectives.

Correspondence to: Paul M. Wechsler, MD, University of Cincinnati, 260 Stetson St, Ste 2300, Cincinnati, OH 45219. Email wechslerpl@ucmail.uc.edu
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To date, work regarding the UW-mRS analysis primarily aimed to derive valid utility weights for the mRS and targeted mainly a clinical trialist audience.^{4,5} As the UW-mRS analysis becomes more common in acute stroke clinical trials, it is critical that the broad clinical and research audiences in the stroke community become increasingly familiar with the methodology of the UW-mRS analysis to ensure clinical trial results are rigorous and interpreted accurately. To contribute to this effort, this narrative review provides a practical overview of key methodology in the UW-mRS analysis to aid in its interpretation, considers the advantages and challenges of using the UW-mRS analysis, and suggests future considerations to refine the UW-mRS analytic approach.

WHAT IS A UTILITY?

Broadly speaking, the term utility refers to the desirability of an outcome to an end-user.⁶ Utilities are used in a variety of fields (economics, business, population health, medicine, etc) to measure the value of different items or outcomes (products, health states) to different end-users (customers, companies, general population, patients, etc). For example, businesses use economic utilities to understand how products or services are valued by their customers. Health economists and health policy experts commonly employ utility weights of health states from different diseases to measure whether treatments, technologies, policies, or other interventions provide value to society based on quality-adjusted life years gained or lost for patients at a certain monetary cost. Utility weighting is commonly applied in clinical medicine to quantify the desirability or valuation of different health states by patients, clinicians, or both. In health applications, utility values typically range from 0.0 (death) to 1.0 (perfect health), but can extend into negative values for health states considered worse than death.

HOW ARE UTILITIES MEASURED?

Utilities in clinical medicine can be elicited from the viewpoint of patients themselves, healthy individuals, or clinicians.

Patients

To elicit utilities from patients, there are 2 broad approaches—direct and indirect. In direct methods, patients directly rate the value of the health state they are currently experiencing. The most common direct method used with patients is the Health Utility Visual Analogue scale.⁷ On a vertical line, anchored by worst imaginable health (0) on the bottom to best imaginable health (100) on the top, patients mark the point that indicates their current health utility. With indirect methods, patients complete generic, multi-attribute health-related quality

of life questionnaires rating their health state in several experiential domains. Responses to the questionnaires are then converted into a utility weight using previously elicited value sets (tariffs) for responses to the questionnaire based on health state preferences previously elicited from large samples of the general population. The most common health-related quality of life rating tool used to do this is the European Quality of Life—5 Dimensions (EQ-5D). The EQ-5D is an internationally used, well-validated survey that assesses patient health-related quality of life in 5 dimensions—mobility, self-care, usual activities, anxiety/depression, and pain.⁸ It is a generic, disease-agnostic tool that can be used to measure health-related quality of life in many disease states. Comprehensive, country-specific value sets for every possible combination of responses to the EQ-5D based on health state preferences previously elicited from samples of the general population are used to convert EQ-5D responses into a single utility weight (Figure 1). This indirect method is the most common, practical, and well-validated approach to eliciting utilities.^{9,10}

Healthy Individuals

In healthy individuals, the most common health state valuation methods are the direct methods of time trade-off and standard gamble. In both, respondents are presented with a description of a disease state, most commonly as a written vignette. In time trade-off, respondents indicate how much shorter a time living in perfect health they would consider equal to living in the disease state. The shorter the time the respondent is willing to live in perfect health compared with living in the disease state, the lower the utility of that disease state. In a standard gamble, they indicate how much risk of immediate death they would accept to have a chance to live in perfect health rather than the disease state.¹⁰ The higher the risk of immediate death they are willing to accept to have a chance to live in perfect health rather than the disease state, the lower the utility for that disease state.

Clinicians

For derivation from clinicians, the most common health state valuation method is the person trade-off. In a typical person trade-off assessment, clinicians are asked to assume they are allocating health system resources and to decide what number N would make them indifferent between the choices of extending the lives of 1000 persons with the disease state for 1 year versus extending the lives of N healthy persons for 1 year.¹¹ This method is commonly used to produce disability weights, which represent the magnitude of health loss associated with an outcome.¹² Disability weights and utilities are similar in that they both quantify health state ratings on a scale

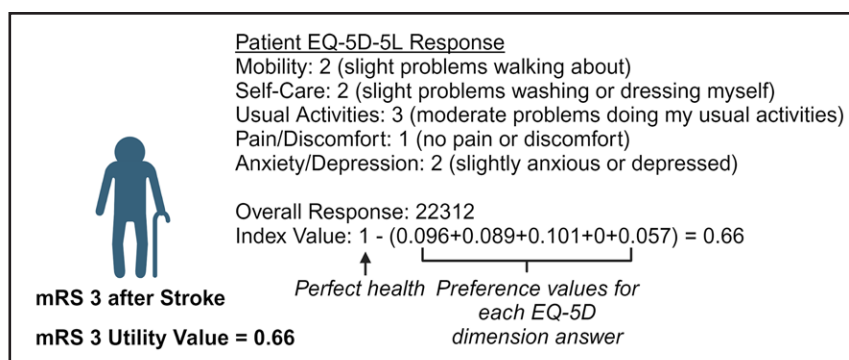


Figure 1. Example of utility weight calculation for a modified Rankin Scale (mRS) level.

A patient who reaches an mRS level of 3 after stroke (able to ambulate independently but dependent on others for some activities of daily living) responds to the EuroQol 5 level (EQ5D-5L) health-related quality of life questionnaire. An index value for that patient's response to the EQ5D-5L is calculated by assigning a health state preference value for their answer in each dimension of the EQ5D-5L based on previously elicited health state preferences for the answers to the EQ5D-5L from a sample of the general US population. This index value serves as the utility weight for an mRS score of 3 for that patient.

from 0.0 to 1.0. However, the disability weight scale is the opposite of the utility scale, with 0.0 representing normal health without disability (no health loss) and 1.0 representing death or a health state as bad as death (all health lost). It is common practice to derive utilities from disability weights by simply taking the inverse ($1 - \text{disability weight}$).^{5,13} A mapping approach to predict utilities from disability weights and vice versa is another methodology but requires more complex statistical modeling.¹⁴

HOW ARE UTILITY WEIGHTS ASSIGNED TO THE MRS?

Several studies have been performed to assign each level of the mRS score (0–6) a health state utility weight representing the desirability or value of that functional status, using different methods and in different populations. The approaches and resulting weightings are shown in the Table.^{5,9,11,15–20}

The most common approach to deriving utility weights for the mRS has been to ask patients with different mRS outcomes to provide multidimensional, health-related quality of life ratings on the EQ-5D and subsequently convert their responses to utility weights using value sets as described in the previous section.^{5,9} For example, functional status measured using the mRS and health-related quality of life using the 3-level EQ-5D survey were assessed at 1, 6, 12, and 24 months after stroke or transient ischemic attack in a population-based British cohort of patients with stroke and transient ischemic attack.⁹ At each time point after stroke or transient ischemic attack, a utility weight for each mRS level was derived by assigning a health state preference value for the patients' EQ-5D responses based on previously elicited health state preferences for the EQ-5D responses from a sample of the general British population (Figure 1).²¹ These patient-derived mRS utility weights were averaged together with utility

weights (converted from disability weights by taking the inverse) derived from an international clinician panel using the person tradeoff method to develop the most widely used mRS utility weights applied in acute stroke clinical trials.^{2,3,5,11,22,23} Although most studies have derived utility weights from general populations and then applied them within clinical trials, a less commonly used approach is to derive utility weights by measuring mRS and EQ-5D for each patient within a specific trial and convert EQ-5D responses into utility weights for each patient in the trial using previously elicited health state preferences for the EQ5D responses from the general population of the country in which the trial is being performed.^{16,18,19} It should be noted that health state preferences for the EQ5D responses elicited from the general population are not available for some countries.

Once it is decided which utility weights for the mRS will be used in a clinical trial, they are applied to the distribution of mRS functional states reached by each group in the trial. Each patient is assigned the utility weight of their mRS outcome level. Treatment group differences in these utility weights can then be compared using statistical models adjusting for baseline covariates of interest.

ADVANTAGES OF THE UW-MRS ANALYSIS

There are several advantages of the UW-mRS analysis that make it an appealing end point in acute stroke clinical trials. Applying patient-derived utility weights to the mRS makes the scale substantially more patient-centered, as opposed to the dichotomized and ordinal approaches to analyzing the mRS that do not consider patient valuations of functional states after stroke. The distinctive, nonlinear variations in patient quality of life between each mRS level are captured by the UW-mRS analysis (Figure 2), making results from acute stroke clinical trials more clinically relevant to patients. There

Table. Examples of Existing Utility Weights for the Modified Rankin Scale

mRS utility weight source	Derivation Approach	Sample size	Utility weight values						
			mRS score, 0	mRS score, 1	mRS score, 2	mRS score, 3	mRS score, 4	mRS score, 5	mRS score, 6
Rivero-Arias et al ⁹	Patient-derived utility weights from a British cohort of patients with stroke and TIA using EQ-5D	2425 patients	1.0	0.87	0.73	0.63	0.31	−0.05	0.0
Hong et al ¹¹	Physician-assigned utility weights (converted to utility weights in this table by the authors of this review by taking the inverse of disability weights) from a panel of neurovascular and emergency medicine physicians from North America and Asia using person-trade-off	18 clinicians	1.0	0.954	0.788	0.669	0.348	0.056	0.0
Chaisinanunkul et al ⁵	Average of utility weights from Rivero-Arias et al and Hong et al Most commonly used mRS utility weights in acute stroke trials. Disability weights from Hong et al converted to utility weights by Chaisinanunkul et al by taking the inverse of disability weights.	2425 patients, 18 clinicians	1.0	0.91	0.76	0.65	0.33	0.0	0.0
Ali et al ¹⁵	Pooled patient-derived utility weights from clinical trial populations from multiple countries. Utility weights reported for each country-specific EQ-5D value set applied to the international patient cohort	4946 patients	0.88–0.95	0.74–0.90	0.51–0.83	0.23–0.70	−0.16 to 0.45	−0.48 to 0.22	0.0
MR CLEAN trial ¹⁶	Patient-derived utility weights from a single clinical trial population of ischemic stroke patients from the Netherlands using EQ-5D	500 patients	0.95	0.93	0.83	0.62	0.42	0.11	0.0
Pinckaers et al ¹⁷	Pooled patient-derived utility weights from 3 clinical trial populations in the Netherlands using EQ-5D	1083 patients	0.950	0.866	0.763	0.599	0.352	0.147	0.0
BEST MSU trial ¹⁸	Patient-derived utility weights from a single clinical trial population of patients with ischemic stroke from the US using EQ-5D	1047 patients	1.0	0.91	0.74	0.65	0.19	0.03	0.0
Wang et al ¹⁹	Patient-derived utility weights from 8 clinical trial populations of patients with ischemic stroke and ICH using EQ-5D	22 946 patients	0.96	0.83	0.72	0.54	0.22	−0.18	0.0
Slaughter et al ²⁰	Patient-derived utility weights from cohort of ICH patients in Texas (US) using direct (standard gamble) and indirect methods (EQ-5D)	280 patients	1.0 (SG)	1.0 (SG)	0.98 (SG)	0.98 (SG)	0.96 (SG)	0.60 (SG)	0.0 (SG)
			1.0 (EQ-5D)	0.86 (EQ-5D)	0.83 (EQ-5D)	0.73 (EQ-5D)	0.41 (EQ-5D)	0.18 (EQ-5D)	0.0 (EQ-5D)

EQ-5D indicates European Quality of Life–5 Dimensions; ICH, intracerebral hemorrhage; mRS, modified Rankin Scale; TIA, transient ischemic attack.

are statistical advantages to utility-weighted analysis of the mRS.⁵ Linear regression can be used to examine differences in average utilities between 2 groups while easily incorporating adjustments for baseline covariates. This modeling requires less rigid assumptions compared with ordinal analysis of the mRS by negating the need to satisfy the proportional odds assumption while maintaining statistical power.⁵ The UW-mRS analysis can capture both the benefit of an intervention at some mRS level transitions and the harm at other transitions between mRS levels. A stroke treatment may improve the rate of patients reaching a favorable outcome after stroke (eg, mRS score, 0–2) while simultaneously resulting in more harm for some patients, which may be reflected in differences in the distribution of functional outcomes across the higher mRS levels (eg, mRS score, 3–6).²⁴ The dichotomized approach does not capture this bidirectional effect as it assesses only 1 health state transition. The mean utilities in the intervention group of a trial using the UW-mRS analysis integrate both beneficial and

harmful effects of the intervention across the distribution of all mRS levels. Lastly, the utilities of stroke treatments generated by using the UW-mRS analysis facilitate the opportunity for health care decision makers to compare the economic and societal value between stroke treatments and interventions for other diseases.

Proponents of dichotomized and ordinal approaches to mRS analysis sometimes suggest these approaches have an advantage over the UW-mRS analysis of making fewer assumptions about each individual patient's valuing of a health state. This is not the case. The dichotomized and ordinal approaches also make assumptions about each patient's valuing of a health state, and their assumptions are highly implausible. Dichotomous analysis of the mRS in effect assigns unrealistic, uniform utility weights to very different health states (Figure 2). For example, dichotomous analysis of mRS score of 0 to 2 versus 3 to 6 effectively assigns a uniform utility weight of 1.0, perfect health, to the 3 very different health states of: mRS score of 0 (no symptoms at all), mRS score of

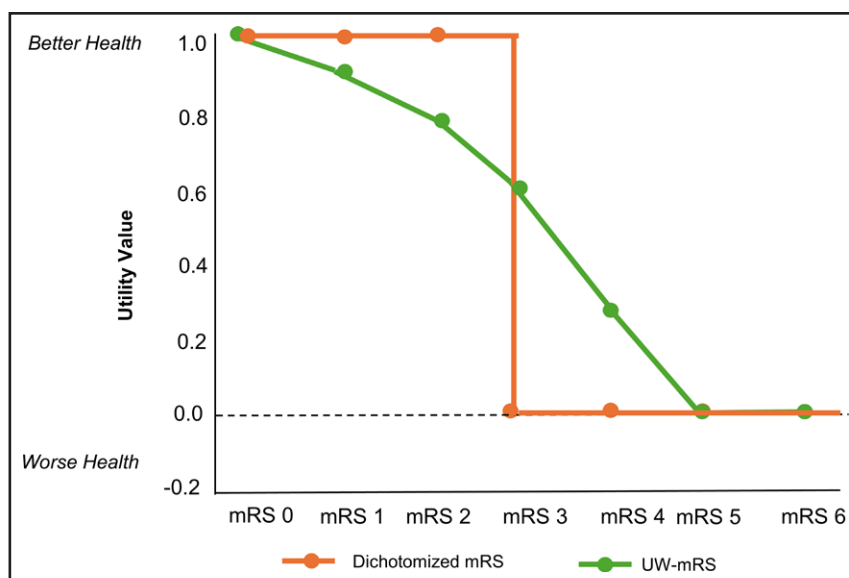


Figure 2. Utility valuation of modified Rankin Scale (mRS) functional status according to how the mRS is analyzed.

A utility value of 0.0 represents death and 1.0 represents perfect health, with negative values representing health states worse than death. The utility-weighted (UW)-mRS analysis captures the actual, nonlinear variation in patient valuation of functional status by quantifying patient health-related quality of life for each mRS level. In contrast, the dichotomized approach makes the simplifying, but clearly false, assumption that several mRS states (eg, 0, 1, and 2) all equally represent perfect health (utility value = 1.0) and the remaining mRS states (eg, 3, 4, 5, 6) all equally represent the worst possible health (utility value = 0.0). The ordinal approach assumes that with increasing levels of disability, each level of the mRS has a lower valuation than the previous level, but nothing can be said about how much lower the valuation of that mRS level is compared with the previous. Thus, specific utility values cannot be assigned for the ordinal approach and cannot be plotted in the figure.

1 (persistent symptoms), and mRS score of 2 (being unable to return to work or fully engage in family and social activities); and also assigns a utility weight of 0.0 to the 4 remaining, very different health states: mRS score of 3 (being unable to live independently), mRS score of 4 (not being able to walk or perform bodily self-care without the help of others), mRS score of 5 (bed bound), and mRS score of 6 (death). The ordinal approach has a different, highly implausible assumption. It assumes that no more can be said regarding mRS valuations other than that they are ordered, without any possible specification of how much 1 level is valued more or less than its neighbors. The broad similarities of mRS level utility weights derived from patients and clinicians across multiple studies belie this assumption (Table).

CHALLENGES AND CONCERNS

Despite its many advantages, concerns have been raised regarding UW-mRS analysis. One critique is that it would be preferable to have each patient assign an individualized utility weight to their mRS outcome rather than using a common utility weight for all patients at each mRS level, since each patient will actually have their own valuation of the mRS level that may differ from the prototypical value of the larger population.¹⁶ However, such individualized patient-reported outcomes are already captured by other outcome measures, such as patient responses to the EQ-5D or other quality of life measurement tools. Using individualized utility weights

would convert the UW-mRS from a distinct outcome designed to reflect population-average valuation of poststroke functional status from the perspective of patients and clinicians to a redundant patient-reported outcome. Another concern is that clinicians may struggle to explain clinical trial results to patients when the UW-mRS analysis was used in the trial. The dichotomized approach to mRS analysis lends itself to straightforward conversion to number needed to treat values for clinical translation to patients and their families—among every X patients treated, 1 more will be independent in taking care of themselves after the stroke. Methods have also been developed to derive estimated number needed to treat values for ordinal mRS outcomes, enabling similar clinical communication—among every X patients treated, one more will have a reduced level of disability after the stroke.²⁵ In contrast, methods have not yet been developed to convert UW-mRS outcome to number needed to treat estimates. It is also important to recognize that the UW-mRS analysis is a measure specifically of disability-related quality of life, not all aspects of health-related quality of life. It can complement but not replace broader and more granular health-related quality of life measures.

Results from an individual trial may differ depending on which approach is used to analyze the mRS.⁵ This is not a challenge unique to the utility-weighted approach when compared with other mRS analytic approaches—differing trial results can also be seen when analyzing trial results using dichotomous versus ordinal analysis.⁵ However,

particular characteristics of the UW-mRS analysis may lend itself to showing different results when compared with ordinal or dichotomized analysis of the same data. Larger or more statistically significant effect sizes may be seen when the UW-mRS analysis is used in trials where treatment benefit is concentrated in changing outcomes in the more disabled mRS levels (mRS score, 3–5). This is because the largest differences in patients' quality of life (utilities) are between mRS scores of 3 and 5 (Figure 2). As an example, the benefit of hemicraniectomy compared with medical management for malignant cerebral infarction was seen most prominently in shifting patients from death to having moderate to severe disability (mRS score, 3–5).²⁶ This results in one of the largest treatment effect sizes seen in stroke trials when using the UW-mRS analysis that is highly statistically significant ($P=0.0001$).⁵ The same results using dichotomized analysis of mRS score of 0 to 2 versus 3 to 6 was still statistically significant favoring hemicraniectomy, albeit to a lesser extent ($P=0.039$), and there was no statistically significant difference when dichotomized mRS score of 0 to 1 versus 2 to 6.⁵ The advantageous ability of the UW-mRS analysis to capture both the benefit of an intervention at some mRS level transitions and potential harm at other transitions between mRS levels can also produce discordant trial results when data is analyzed using other mRS analytic methods. Analysis of the ECASS III trial is an illustrative example.²⁴ More patients who received alteplase for treatment of acute ischemic stroke had an excellent outcome (mRS score, 0–1), but there was no statistically significant difference in outcomes when analyzed using the UW-mRS approach.⁵ This may be partially due to the observation that there were numerically more patients in mRS levels 5 and 6 combined (both with utilities of 0) in the alteplase group compared with the placebo group.²⁴ These features of the UW-mRS analysis that may result in different conclusions to be drawn from the same data when analyzed using the dichotomized or ordinal approach should be kept in mind when designing and interpreting trials.

CONSIDERATIONS OF MRS UTILITY WEIGHTS DERIVED FROM DIFFERENT POPULATION SOURCES

There are advantages and disadvantages of mRS utility weights derived from different populations. Patients themselves are, of course, the most important source of valuations, based on their lived, subjective experience. When patients with stroke cannot participate in the utility weight elicitation process, proxy responses from family or caregivers who are familiar with the patient's values are often utilized. This allows for the mRS utility weights to be more representative of poststroke health state valuation for the real-world stroke population, of which up

to an estimated 60% have communication or cognitive impairments poststroke that may inhibit their ability to participate in utility weight elicitation.²⁷ However, healthy individuals who do not directly experience a health state or personally suffer from a disease tend to overestimate disability and underestimate quality of life compared with individuals actually living in that health state.²⁸ This issue of proxies rating patient outcomes worse than patients rating their own outcomes certainly needs to be recognized, particularly for more subjective domains of health-related quality of life.²⁹ However, restricting utility weight derivation only to patients who can self-respond would limit the generalizability of trial results that use those resulting utility weights. The advantage of including clinician ratings arises from their clinical experience seeing a breadth of functional outcomes after stroke—they observe the patient experience across all levels of the mRS and so can provide a broad perspective on the value of all mRS levels. Patients, in contrast, generally only experience one mRS functional state, though they do so in depth. An advantage of the most used mRS utility weights is that they were derived by combining the perspectives of both patients and clinicians, and the weights from both sources were very similar.⁵

Using fixed mRS utility weights derived from populations outside of a clinical trial and applying them to the distribution of mRS within the trial of interest is the most common method used in acute stroke clinical trials.^{2,22,23} This method improves the generalizability of trial results by reflecting the valuation of poststroke functional status from a broader stroke population. The alternative approach of deriving utility weights within individual trial populations in which the trial was conducted limits external validity, as clinical trial populations are commonly not well representative of populations encountered in routine clinical care.³⁰ Additionally, the process of deriving utility weights within an individual trial involves collapsing health-related quality of life data (commonly EQ5D) into mRS functional outcome data, thereby reducing the granularity of quality of life outcome data.

Demographic, clinical, economic, and sociocultural characteristics of individuals within the population from which utility weights are derived may affect valuation of poststroke health states and mRS utility weights. Valuation of health states may vary widely between countries due to a variety of reasons, including differences in demographics, extent of comorbid diseases contributing to disability and impairments, degree of family and governmental support for living with disability, and sociocultural differences in the acceptability of social role change, pain, and expression of emotions.^{15,31} The most commonly used mRS utility weights incorporate perspectives from patients or clinicians from Britain, the United States, South Korea, China, and Taiwan.^{9,11} This is important as acute stroke clinical trials are increasingly being conducted globally. Although these mRS utility weights

do have international perspectives, they do not include perspectives from large populations from which acute stroke clinical trials are commonly conducted, including mainland European countries, Canada, and Australia.

Specific neurological impairments that are present in a stroke patient population could affect mRS utility weighting. For example, the patient with hemi-spatial neglect who fails to recognize the existence or severity of limb weakness may value their functional status differently compared with the patient with full awareness of these same deficits and functional limitations. Neurovascular injury to specific brain areas may alter an individual's emotional regulation, a factor known to affect health-related quality of life and thus utility valuation.³²

The amount of time spent with impairments or experiencing a certain state of health also affects an individual's valuation of that health state.³³ Patients can physically, psychologically, and socially adapt to their impairments and disability over time. In a majority of patients, illness acceptance and reframed perceptions of social roles lead to valuing the same disabling health state more highly after being in that state for months or years compared with during its initial onset.^{33,34} This response-shift phenomenon may result in an individual assigning different utility weights to the same mRS functional status depending on at what time point after stroke the utility weights are being measured (Figure 3). Relatedly, different functional recovery trajectories may affect valuation of poststroke functional status at a specific time point. An individual who recently transitioned to being ambulatory (mRS score, 3) after several months of being unable to walk may value that new functional status differently compared with an individual who has been ambulatory for many months. Both individuals are in an mRS score of 3 state at the same poststroke time point, but the individual who recently gained the ability to walk without assistance may value their new functional status higher (Figure 3). The studies that have derived utility weights for the mRS generally did not consider alterations in valuation related to types of impairments, at what time point after stroke utility weights for the mRS were derived, and the trajectory of the patient's arrival at their mRS state.

FUTURE DIRECTIONS

We consider the topics below that could help refine the UW-mRS analytic approach for it to be a methodologically robust acute stroke clinical trial end point.

Rethinking Methods to Elicit mRS Utility Weights

There is no universally accepted method that is the standard for eliciting utility weights for the mRS. Eliciting utility weights involves deciding who the utility weights should be elicited from, which measurement tools to

use, and at what time point poststroke utility weights are elicited.

More studies are needed to derive mRS utility weights that reflect patient valuation of poststroke functional status from representative populations of stroke patients in multiple countries. Ideally, these studies would be population-based or large multicenter studies that reflect the full breadth of stroke patients encountered in routine clinical practice in each country. Future studies deriving representative utility weights would be particularly useful in countries where many acute stroke trials are conducted (United States, Canada, Europe, China, and Australia). This would allow country-specific utility weights to be applied when a trial is conducted in 1 country, while facilitating more accurate aggregation of utility weights from multiple countries to be applied in multinational acute stroke clinical trials.¹⁹ It would also be desirable, in countries where utility weight derivations have been made, to regularly perform studies updating the weights to reflect interval sociocultural and socioeconomic changes.

Future studies deriving mRS utility weights should consider the measurement tools used to elicit the utilities. Current mRS utility weights used the generic EQ-5D questionnaire due to its ability to generate utility weights that can be compared with other disease states and the convenience of using existing value sets for the responses. However, the EQ-5D only probes domains affected by stroke in a cursory manner, for example, with only a single question each for mobility, self-care, and usual activities. Future studies could consider deriving utility weights using quality of life questionnaires that are specific to stroke and probe the patient experience of stroke-related disability in a much more detailed manner, such as the Stroke Impact Scale.³⁵ This would require further research to develop standardized value sets for the responses to those questionnaires. Another target for future studies is to address the limitation that, to date, mRS utility weights have been based on EQ-5D value sets derived from asking members of the general public to value a set of hypothetical given health states described to them. An alternative is to develop value sets for the EQ-5D using direct methods by asking actual patients to value only the health state they are currently experiencing, an approach referred to as experienced health states.³⁶ Experienced health states derived value sets can differ substantially from given health states derived value sets.³⁶ Although experienced health states health state value sets have been derived based on collated assessments from patients with different chronic diseases, including stroke, valuations based solely on patients with stroke are not widely available. Future studies should consider using direct methods to elicit the EQ-5D value sets directly from stroke patients to avoid projecting potentially inaccurate, hypothetical health valuations from the general population.²⁰

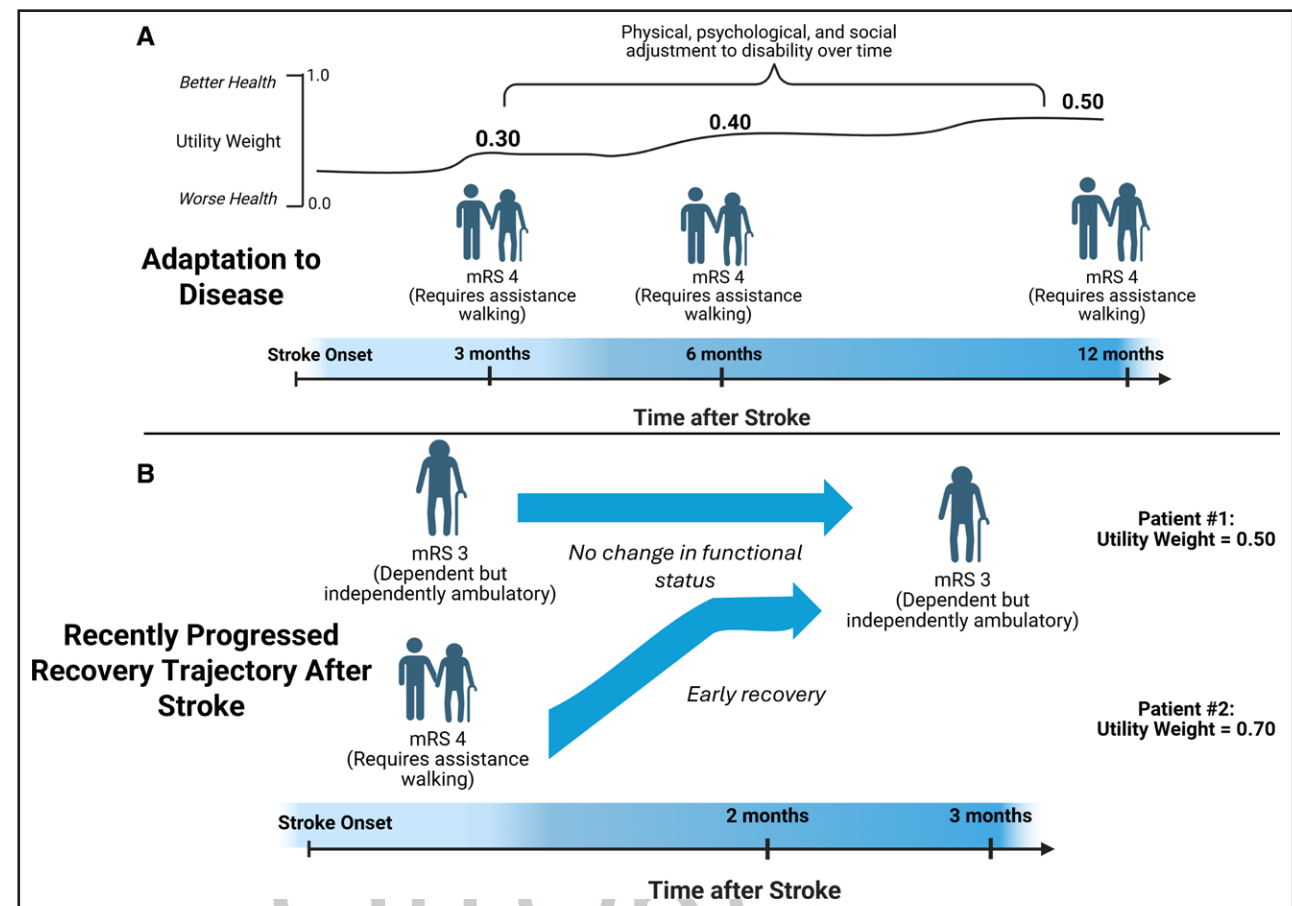


Figure 3. Potential effects of time on modified Rankin Scale (mRS) utility weights.

A, Patients can physically, psychologically, and socially adapt to impairments and disability over time. Illness acceptance and reframed perceptions of social roles lead to valuing the same disabling health state more highly after time has passed (response-shift phenomenon). **B**, Different recovery trajectories over time to finally reach an mRS level may influence the values patients assign to the mRS level. In this example, 2 patients both have mRS 3 (dependent on others for some activities of daily living) at 3 months after stroke. However, the first patient spent the entire 3 months poststroke at an mRS score of 3, while the second patient recently transitioned from mRS 4 (requires assistance walking) to an mRS score of 3. The second patient may value their new level of functioning with less disability compared with the patient who has been living in the mRS score of 3 functional status for longer.

The currently available mRS utility weights do not explicitly consider at what time point poststroke utility weights were derived. The most commonly used patient-derived mRS utility weights were not elicited at 3 months poststroke,⁹ which is the most common time point acute stroke trials measure functional outcomes. Thus, due to the adaptation to disease and disease trajectory effects of time on health state valuation, it is possible that currently available mRS utility weights do not accurately reflect patient valuation of functional status at the most common time point used in acute stroke trials. Additionally, for more severe stroke conditions that take a longer time to recover and to assess durability of 3-month effects than in more moderate stroke conditions, acute stroke clinical trials increasingly measure functional outcomes beyond the traditional 3-month poststroke time point.^{23,37} Future studies measuring utility weights for mRS levels at different time points poststroke would be beneficial so that trials could use utility weights corresponding to

specific time points poststroke depending on when after stroke the mRS is measured in the trial.

Translating the UW-mRS to the Bedside

What does a 0.2 higher utility weight from a stroke treatment compared with control mean to the patient in front of a clinician? The answer to this is not immediately apparent. Although the UW-mRS analysis improves the patient-centeredness of the mRS, its use paradoxically makes translating trial results to patients more difficult. Addressing this issue will require a multicomponent approach. Efforts such as this review are needed to familiarize clinicians with what the utility weights represent and how they are derived, increasing understanding of UW-mRS analysis result values. Qualitative studies with multidisciplinary stakeholders including clinicians, caregivers, and patients are needed to determine the most effective content and wording that improves patient

health literacy of trial results that use the UW-mRS analysis. Lastly, more research is needed to understand the minimally clinically important difference in utilities for patients with stroke. Current benchmarks for minimally clinically important difference of the EQ-5D (from which mRS utility weights are derived) were generated from nonstroke populations.^{38,39} These range from 0.03 to 0.10, which is relatively consistent with the average differences in utilities between groups in re-analyses of acute stroke clinical trials using the UW-mRS analysis.⁵ Rigorous studies should confirm whether these thresholds hold true for patients with stroke to ensure acute stroke trials are powered adequately to detect clinically meaningful results.

Opportunity to Include Patients With Prestroke Disability in Trials

UW-mRS analysis may facilitate opportunities for trialists to include in acute stroke trials patients who had already been living with disability before onset of the index stroke (Figure 4). Despite constituting nearly one-third of the stroke population and one-fourth of the US population,^{40–42} patients with restrike disability (mRS score, 2–5 before stroke) are commonly excluded from acute

stroke clinical trials.^{43–46} This is because of limitations in the common approaches to analyzing the mRS. In the dichotomized approach that uses excellent (mRS score, 0–1, able to do work/leisure/school activities full time) or good (mRS score, 0–2, able to live independently) disability thresholds to define favorable outcomes from a stroke intervention, favorable poststroke outcomes are often unattainable by patients who had an established prestroke disability (mRS score, 2–5) or dependency (mRS score, 3–5).¹ In the ordinal approach, although patients with prestroke disability can contribute to ordinal group differences, there is limited statistical power in analyzing differences in the distribution of the mRS between 2 groups because outcomes will be distributed over only a subset of mRS levels, given patients with prestroke disability start in higher mRS levels before stroke (mRS score, 2–5). Using the UW-mRS analysis could allow for patients with prestroke disability to be included in trials because all patients, regardless of whether disabled before stroke or not, can contribute to the average utility value for the group without loss of statistical power (Figure 4). The largest differences in patients' quality of life (utilities) are between mRS scores of 3 and 5 (Figure 2), which are all functional states considered disabled by clinicians. Therefore, using the UW-mRS

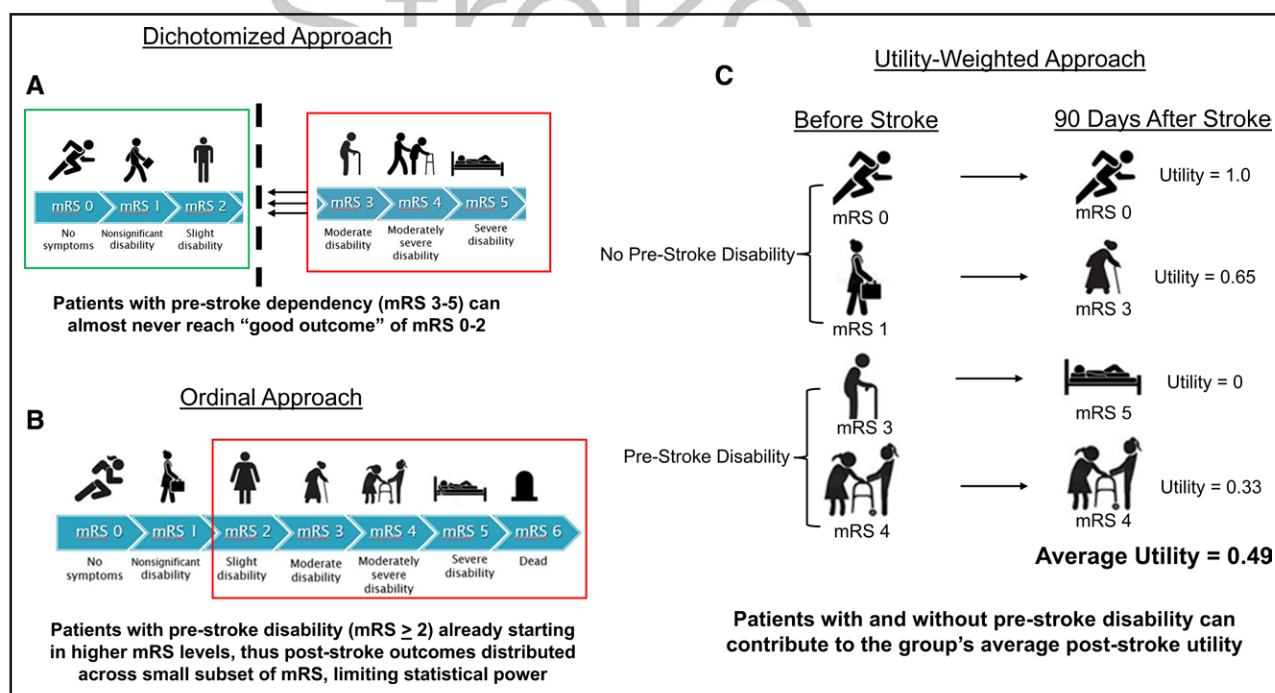


Figure 4. Utility-weighted modified Rankin Scale (UW-mRS) analysis facilitates the opportunity to include patients with prestroke disability in acute stroke clinical trials.

A, In the dichotomized approach, a patient with prestroke dependency (mRS score, 3–5 before stroke) can never reach a good outcome of mRS score of 0 to 2 (able to live independently), and their course is uninformative about treatment effect when dichotomous analysis is used. **B**, In the ordinal approach, given that patients with prestroke disability start in higher mRS levels before stroke (mRS score, 2–5), there is reduced statistical power in analyzing differences in the distribution of the mRS between 2 groups because outcomes will be distributed over only a subset of mRS levels. **C**, UW-mRS analysis allows patients with prestroke disability to be included in acute stroke trials because even if they reach a disabled functional status, they can contribute to the group's average utility. Adapted from Saver et al,¹ with permission. Copyright ©2026, Wolters Kluwer.

analysis to investigate a stroke treatment that may prevent a patient who needs assistance preparing a meal (mRS score, 3) from becoming bedridden (mRS score, 5) may improve patient quality of life quite substantially and be a highly effective treatment from the patient's perspective. Future research is needed to derive mRS utility weights from populations of patients who were disabled before stroke to understand how patients with prestroke disability value their poststroke functional status, if this valuation differs from patients without prestroke disability, and which tools are best suited to capture health state valuation in this population.

CONCLUSIONS

UW-mRS analysis holds great promise in improving the patient-centeredness of the mRS. By capturing how patients value their functional status after stroke, UW-mRS analysis moves a step closer to capturing the true lived experience of patients suffering from stroke. Future work is needed to determine the best methodology for eliciting mRS utility weights, which populations to elicit these from, how to further incorporate diverse perspectives of the value of health states in multinational acute stroke trials, and how to best communicate trial results that used the UW-mRS analysis to patients.

ARTICLE INFORMATION

Affiliations

University of Cincinnati, OH (P.M.W., J.P.B., E.A.M.). University of California Los Angeles (J.L.S.).

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